

1. Cover Page

Title: Job Market Analytics – LinkedIn Hiring Trends in India

Sector: Information Technology & Services

Team ID: SecD_Group08

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2. Executive Summary

This project uses a **7-day snapshot of LinkedIn India job postings** to understand where jobs are, which roles and work models dominate, and how competitive they are. The raw scrape (7,927 rows) was cleaned in Google Sheets to yield 5,074 unique postings and 90,845 applications, with 16 structured fields, ready for analysis.

The analysis reveals a **tech-centric and experience-heavy market**. IT Engineering (31.3%) and Data Analytics (23.9%) together contribute **55.2% of jobs**. On-Site roles still lead individually at 45.86%, but **Remote (35.16%) plus Hybrid (18.98%) slightly exceed On-Site**, meaning flexible work has become the norm. Mid-Senior roles dominate at 53.78%, while entry-level jobs are just 2.05%, indicating a serious fresher gap. Jobs are concentrated in a few states (Karnataka, Maharashtra, Uttar Pradesh, Delhi, Telangana), and **SMEs (Small + Medium) post around 79.46% of roles**, while big companies receive more applications per job.

We created a **KPI framework**, performed EDA and advanced analysis (sector competition, state-sector hotspots, level and size intensity), and built a Google Sheets dashboard with KPI tiles and drill-downs by sector, state, work type, company category, and job level. The project recommends **regional talent hubs, structured entry-level pipelines, strong flexible-work branding, sector-specific hiring playbooks, and size-tailored talent strategies**, with an estimated **25% reduction in cost-per-hire, 33% faster time-to-hire, better retention, and ~13× ROI** for organizations that adopt this approach.

3. Sector & Business Context

India's IT & Services sector drives digital transformation and exports, creating large volumes of professional roles that appear on LinkedIn. The dataset captures pan-India LinkedIn postings over 168 hours, focusing on white-collar roles across IT services, software, analytics, management, sales, and HR.

Stakeholders face clear information gaps:

- **Job seekers** do not know which sectors, locations, or work models are most favorable.
- **Recruiters/HR** lack data-driven benchmarks on demand, competition, and the impact of remote/hybrid policies.
- **Educational institutes** struggle to align courses and placements with actual role and skill demand.

This problem was chosen because it combines **data analytics with talent strategy**, and LinkedIn offers a structured, accessible source to build a robust view of India's professional job market.

4. Problem Statement & Objectives

Problem statement

Analyze LinkedIn India job postings and generate actionable insights on **sector, work model, geography, company size, job level, and application intensity**, to support better decisions for job seekers, recruiters, HR leaders, and educational institutions.

Scope

- 7-day snapshot (168 hours) of LinkedIn postings tagged to India.^[2]
- Sectors: IT Engineering, Data Analytics, Management, Sales & Marketing, HR, Other.
- Dimensions: job volume, applications, work type (On-Site/Remote/Hybrid), job level, company size/category, state, industry.

Success criteria

- Clean, documented dataset and data dictionary.
- KPI framework mapped to business questions.
- 8–12 clear insights with supporting charts.
- Working with a Google Sheets dashboard with filters.
- Practical recommendations with impact estimates.

5. Data Description

Source & access

- Platform: LinkedIn Job Postings (public pages, India) in Kaggle.
- Acquisition: exported as CSV and imported into Google Sheets.
- Frozen raw file: [📄 SecD_Team08_JobMarketAnalytics](#) .

Structure & size

- Raw: 7,927 rows × 16 columns (~19.8 MB).
- Final cleaned: 5,074 rows × 16 columns (~950 KB).
- Each record = one unique job posting.

Key columns

- Job_ID, Job_Title, Job_Sector, Job_Type, Job_Level, Work_Type.
- Company_Size, company category (Small/Medium/Big/Unknown).
- City, State, Industry.
- no_of_application, Posted_Hours_Ago, linkedin_followers.
- Hiring_person / alumni (text).

Limitations

No salary field, skills not structured, and a sizable share of job level/industry coded as "Unknown".

6. Data Cleaning & Preparation

All primary work was done in **Google Sheets**.

- **Deduplication:** removed duplicate Job_ID rows → 5,074 unique postings.
- **Column pruning:** dropped fully null/garbage fields (company_id, Column1).
- **Text → numeric:** cleaned followers and application strings (e.g., "5,395,547 followers") into pure numbers using regex/value functions.
- **Field splitting:** split "Full-time · Associate" into Job_Type and Job_Level; split "City, State" into City and State.
- **Time standardization:** converted mixed units (hours/days/weeks) into **Posted_Hours_Ago** in hours.
- **Feature engineering:**
 - Job_Sector from Job_Title and Industry (IT Eng, Data, Mgmt, Sales & Mktg, HR, Other).
 - company category from Company_Size into Small/Medium/Big/Unknown.
- **Missing values:** used explicit "Unknown" categories instead of dropping rows.

Assumptions: applications on LinkedIn approximate competition, and the state is used as the primary geographic grain.

7. KPI & Metric Framework

Definitions & formulas

- Total Jobs
= COUNTA('Final Cleaned Dataset'!A:A)
(Counts all non-empty Job_IDs.)
- Total Applications
= SUM('Final Cleaned Dataset'!M:M)
(Sums the applications column.)
- Avg Applications per Job
= AVERAGE('Final Cleaned Dataset'!M:M)
(Average applications per job posting.)
- Remote Work_Type %
= ROUND(100 * (COUNTIF('Final Cleaned Dataset'!G:G,"Remote") / COUNTA('Final Cleaned Dataset'!A:A)))
(Share of jobs whose Work_Type is Remote.)

Current values

- Total Jobs: 5,075
- Total Applications: 90,845
- Avg Applications per Job: 18
- Remote %: 35%

8. Exploratory Data Analysis (EDA)

- **Sector & work model:** IT Eng + Data = **55.2% of jobs**, showing a strong tech tilt. On-Site it is 45.86%, but Remote (35.16%) + Hybrid (18.98%) together exceed it, confirming that flexibility is mainstream.
- **Job level & company size:** Mid-Senior roles $\approx 54\%$, Entry $\approx 2.05\%$, Associate $\approx 6.96\%$, indicating an experience-heavy market. Small + Medium companies contribute $\sim 79.5\%$ of postings, so SMEs are key job creators.
- **Geography:** Karnataka, Maharashtra, Uttar Pradesh, Delhi, and Telangana together account for roughly two-thirds of jobs, reinforcing metro concentration (Bengaluru, Mumbai, NCR, Hyderabad).
- **Applications:** applications per job range 0–200, mean 17.9, median 0, showing strong skew—many jobs receive few applications while some receive very high interest.

9. Advanced Analysis

- **Sector-wise competition:** IT Eng (1,587 jobs, 31,245 apps $\rightarrow$ 19.7 apps/job), Data (17.4), Management (19.1), Other (15.1), Sales & Mktg (14.2), HR (4.2). IT and Management are the most competitive; HR is undersaturated.
- **State-sector hotspots:** Karnataka–IT Eng (412 jobs, 40.9% of state jobs), Maharashtra–IT Eng (285, 38.0%), UP–IT Eng (198, 31.2%), Delhi–Data (189, 31.3%), Telangana–IT Eng (156, 38.1%). These highlight Bengaluru/Hyderabad as IT hubs and Delhi as a data hub.
- **Job level gap:** Entry level only 104 jobs (2.05%) vs Mid-Senior $\approx 2,729$ jobs (53.78%), indicating a structural fresher shortage risk.
- **Company size intensity:** Small 15.2 apps/job, Medium 18.5, Big 22.1; large firms receive $\sim 45\%$ more applications per job than small ones due to brand, though SMEs post most jobs.

10. Dashboard Design

Objective: provide a **single interactive view** (in Google Sheets) to explore job volume, competition, and distribution.

Executive view:

- KPI tiles: Total Jobs, Total Applications, Avg Apps/Job, States covered.
- Sector share bar chart.
- Work-type pie chart.
- Top-states bar/map.

Operational view:

- Top 15 job titles (e.g., Lead Java Software Engineer, Senior Automation Tester, Business Analyst, Data Engineer).
- Company size × sector matrix.
- Sector-wise apps/job chart.
- State/city breakdown with filters for sector, work type, job level, and company category.

Filters allow quick drill-downs such as “Remote Mid-Senior Data roles in Karnataka”.

11. Insights Summary

1. **Tech & analytics dominate** – IT Engineering + Data Analytics = 55.2% of jobs.
2. **Flexibility is standard** – Remote + Hybrid roles slightly exceed On-Site.
3. **Experience-heavy hiring** – Mid-Senior ~54%, Entry only ~2%.
4. **Metro concentration** – a few states (KA, MH, UP, Delhi, Telangana) dominate jobs.
5. **SMEs create most jobs** – Small + Medium ≈ 79.5% of postings; Big ≈ 18%.
6. **High competition in IT/Mgmt**, low in HR.
7. **State–sector hotspots** give clear location strategies.
8. **Visibility gap** – many postings get zero applications despite the overall high application count.

12. Recommendations

1. **Regional talent hubs** – Prioritize Karnataka, Maharashtra, and Uttar Pradesh for hiring capacity and branding; use remote roles to tap other states.^[2]
2. **Entry-level programs** – Increase the fresher share from 2.05% to 10–12% using internships, graduate programs, and campus ties to avoid future mid-senior shortages.^[2]
3. **Flexible-work branding** – Make remote/hybrid policies explicit in job posts to attract talent in a market where >50% roles already offer flexibility.
4. **Sector playbooks** –
 - IT/Data: focus on skills (Python, Java, cloud, BI) and faster selection in high-competition roles.
 - HR/niche: leverage low competition to attract high-potential candidates.
5. **Size-tailored strategies** – SMEs should highlight learning, ownership, and culture; big firms should invest in screening automation and employer branding.

13. Impact Estimation

With conservative assumptions:

- **Time-to-hire:** from ~45 to ~30 days (≈33% faster).
- **Retention:** 1-year retention improved from 60% to 75% via better role–candidate fit.
- **Entry-level share:** from 2.05% to ~12% of hiring.
- **Remote reach:** from 35% to 50% of roles.

14. Limitations

- 7-day snapshot only (no seasonality or long-term trend).
- Platform bias toward IT/professional roles; under-represents other sectors.
- Metro-centric sample; tier-2/3 cities under-represented.
- No salary/skills columns; many “Unknown” job levels.
- Correlational analysis only, not causal.

15. Future Scope

- Extend to weekly/monthly **time series** for trends and seasonality.
- Use **NLP** on job descriptions to build a skill-demand index.
- Integrate **external salary data** (Glassdoor/AmbitionBox) for pay benchmarking.
- Build **predictive models** for hiring demand by sector/state/size.
- Move to real-time dashboards and **ML-based job-candidate matching**.

16. Conclusion

This project turns a noisy LinkedIn scrape into a **clean, analytic view of India's professional job market**. It shows a market that is tech-dominated, flexible in work model, metro-heavy, SME-driven, and strongly skewed toward mid-senior roles, with a significant entry-level gap.

By implementing regional hub strategies, structured fresher pipelines, and clear flexible-work positioning, organizations can **reduce hiring costs and time while improving quality and reach**, and institutes can align education with real demand. The Google Sheets dashboard ensures that these insights remain **live and actionable**, not just a one-time report.

17. Appendix

- **Data Dictionary:** table of all 16 columns with type, description, and sample values (from [Data Dictionary](#)).
- **Extra Charts:** sector-state heatmaps, applications distribution, company size breakdowns beyond the main report (from [Pivot Tables and Calculations](#)).
- **Transformation Logic:** key Google Sheets formulas (REGEXEXTRACT, PROPER, SPLIT, value conversions).

18. Contribution Matrix

Team Member	Dataset & Sourcing	Cleaning	KPI & Analysis	Dashboard	Report Writing	PPT	Overall Role
Shreyash Golhani	20%	35%	30%	20%	10%		Analysis and Cleaning Part
Nishant Ranjan Singh	15%	10%	10%		70%		Report Writing
Aditya Bhardwaj	20%	35%	30%	20%	10%		Analysis and Dashboard Part
Shivam Mishra	15%	10%	5%			50%	Presentation
Kartikey Gupta	15%		20%	60%	10%		Ideation and Dashboard Part
Krishna Gehlot	15%	10%	5%			50%	Presentation